

# Vol 12 Chapter 4 — Chemical Reactions and Thermodynamics

Keeper (author pass — deep math/physics revision)

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## Chapter 4 — Chemical Reactions and Thermodynamics

*Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine\_DIV5\_QM\_GR\_SM/) or the theorem registry disagree, they win.*

### Level 1 — one sentence

Chemical reactions are substrate K-type rearrangements with thermodynamic state functions ( $\Delta_r H$ ,  $\Delta_r S$ ,  $\Delta_r G$ ) governing direction and equilibrium —  $\Delta G^\circ < 0$  spontaneous; equilibrium constant  $K = \exp(-\Delta G^\circ/RT)$  — and BST views the whole process as substrate count-maximization in chemical-K-type configuration space.

### Level 2 — graduate-physicist precision

#### 4.1 Thermodynamic state functions

For a reaction  $aA + bB \rightarrow cC + dD$ : - Enthalpy  $\Delta_r H = \sum \text{products} - \text{reactants}$  (heat at constant P) - Entropy  $\Delta_r S = \Delta \text{entropy change (count of microstates)}$  - Gibbs free energy  $\Delta_r G = \Delta_r H - T\Delta_r S$  (driving force at constant T, P)

Spontaneous:  $\Delta_r G < 0$ . Equilibrium:  $\Delta_r G = 0$ .

#### 4.2 Equilibrium constants

For ideal-gas reaction at constant  $T$ :

$$K_p(T) = \exp(-\Delta G^\circ/RT)$$

with  $K_p = \prod (P_i/P^\circ)^{\nu_i}$  over products / reactants with stoichiometric coefficients  $\nu_i$ .

Le Chatelier: equilibrium shifts to counteract perturbations (add reactant  $\rightarrow$  forward; raise T for endothermic  $\rightarrow$  forward; etc.).

#### 4.3 Standard reduction potentials

Half-cell reactions  $\text{Ox} + ne^- \rightarrow \text{Red}$  have standard reduction potentials  $E^\circ$  (relative to SHE = standard hydrogen electrode).

Cell potential:  $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ .

Nernst:  $E = E^\circ - (RT/nF) \ln Q$  — non-equilibrium concentrations.

Driving voltage of batteries, electrolysis, biochemical electron transport.

#### 4.4 Examples

Hydrogen combustion:  $H_2 + (1/2)O_2 \rightarrow H_2O$ .  $\Delta_r H = -286$  kJ/mol (highly exothermic).  $\Delta_r G = -237$  kJ/mol. Drives fuel cells.

Photosynthesis:  $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ .  $\Delta_r G = +2870$  kJ/mol (non-spontaneous; driven by photon energy).

Glycolysis: glucose  $\rightarrow$  2 pyruvate + 2 ATP + 2 NADH. Net  $\Delta G \approx -146$  kJ/mol; couples thermodynamics to biological energy storage.

#### 4.5 Substrate-mechanism reading

Per Casey's Principle (Vol 6 Ch 1): entropy = force = counting at depth 0. Chemical reaction directions are substrate's K-type count gradients; equilibrium = substrate count-maximum given constraints.

#### 4.6 K-audit anchors

- Vol 6 Ch 1-4: thermodynamics foundation
- Casey's Principle (memory)

### Level 3 — 5th-grader accessibility

Chemical reactions: bonds break, new bonds form. Thermodynamics tells you direction:  $\Delta G = \Delta H - T\Delta S$  — if  $\Delta G < 0$ , reaction goes forward. Equilibrium constant  $K = \exp(-\Delta G^\circ/RT)$ . Le Chatelier: perturb equilibrium, it shifts to counteract. Standard reduction potentials: drive batteries and biology. In BST: reactions are substrate K-type rearrangements; thermodynamics is substrate counting.

What comes next

Chapter 5 develops reaction kinetics and transition states.

Where to look this up

- Atkins, *Physical Chemistry*
- BST: Vol 6 Ch 1-4